

Name: _____

SOLUTIONS

Date: _____

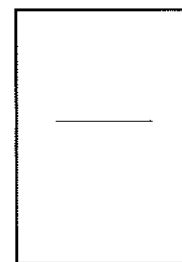


Year 9 Ace/Maritime/General Mathematics

Investigation 4, 2015

Topic – Transformation of Quadratic Equation

Take Home Component



Total Time: *60 minutes*

Reading Time: *N/A*

Working Time: *N/A*

Weighting: *7% of the year.*

Equipment: *Calculator*

Date out: *Week _____ Date _____* **Date Due:** *Week _____ Date _____*

Take home component weighting: *0% of the year*

In-class component weighting: *7% of the year.*

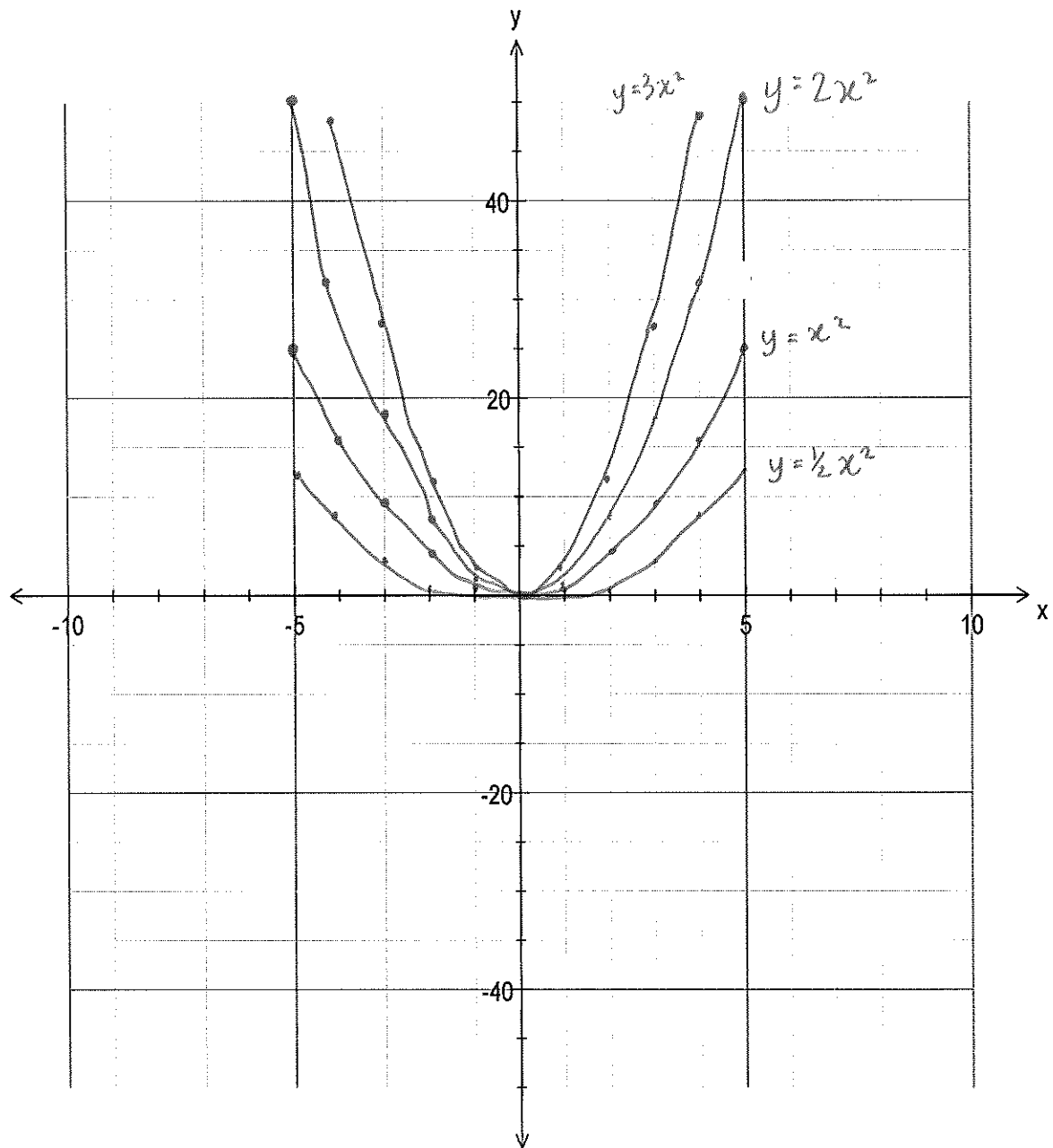
AIM: *In this assessment, you will be investigating the transformation of Quadratic Equation.*

1.

a) Complete the table of values for the equations shown.

	-5	-4	-3	-2	-1	0	1	2	3	4	5
$y = x^2$	25	16	9	4	1	0	1	4	9	16	25
$y = 2x^2$	50	32	18	8	2	0	2	8	18	32	50
$y = 3x^2$	X	48	27	12	3	0	3	12	27	48	X
$y = 0.5x^2$	12.5	8	4.5	1	0.5	0	0.5	2	4.5	8	12.5

b) Now draw each of the equations on the axis below, be sure to label each equation.



c) Looking at the graphs you have drawn in the form $y = ax^2$, describe how the value of "a" changes (transforms) the original equation $y = x^2$?

The higher the value of a the narrower it gets.

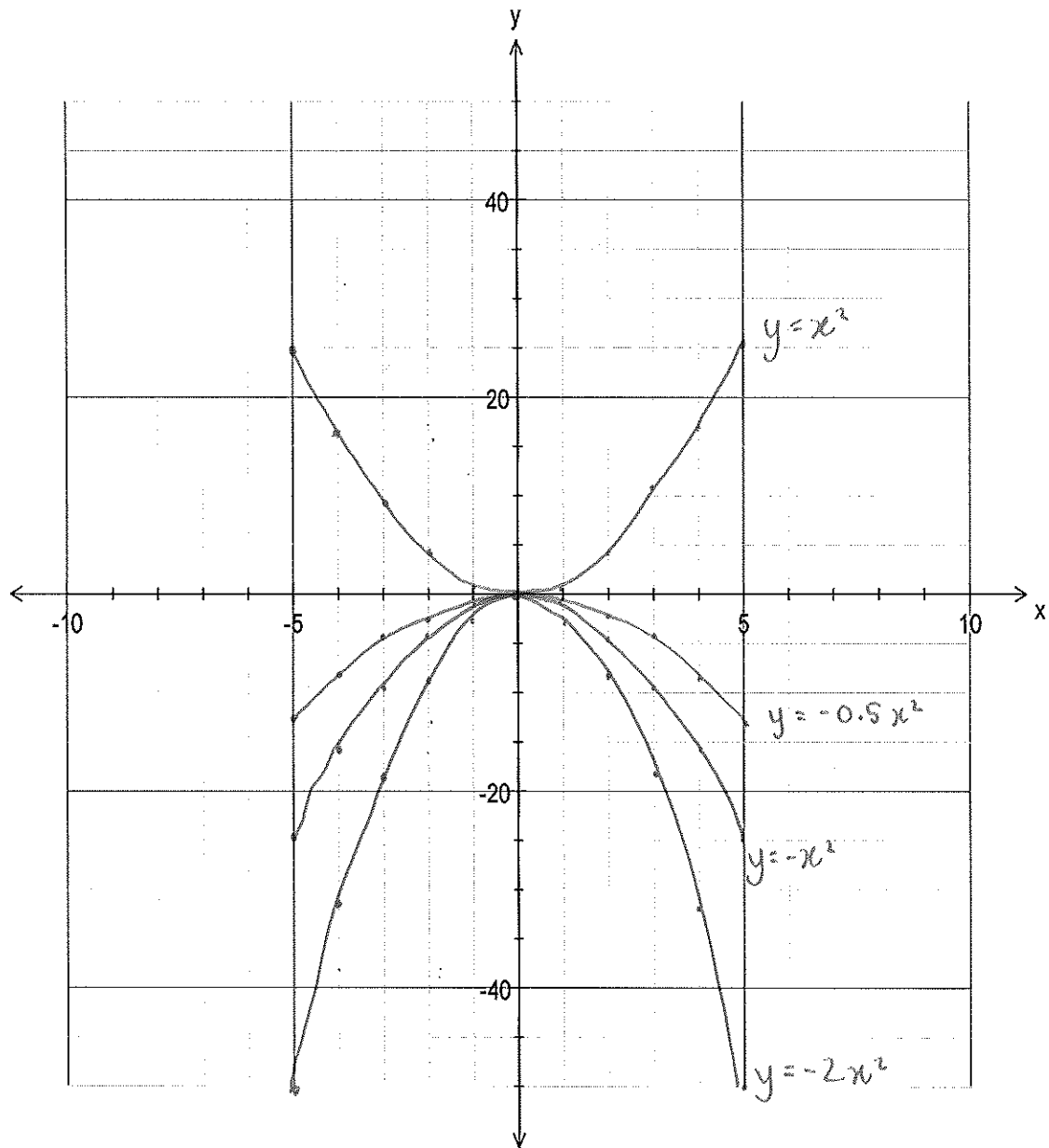
A fraction value of a made it wider.

2.

a) Complete the table of values for the equations shown.

	-5	-4	-3	-2	-1	0	1	2	3	4	5
$y = x^2$	25	16	9	4	1	0	1	4	9	16	25
$y = -x^2$	-25	-16	-9	-4	-1	0	-1	-4	-9	-16	-25
$y = -2x^2$	-50	-32	-18	-8	-2	0	-2	-8	-18	-32	-50
$y = -0.5x^2$	-12.5	-8	-4.5	-2	-0.5	0	-0.5	-2	-4.5	-8	-12.5

b) Now draw each of the equations on the axis below, be sure to label each equation.



c) Looking at the graphs you have drawn in the form $y = ax^2$, describe how the value of negative "a" changes (transforms) the original equation $y = x^2$?

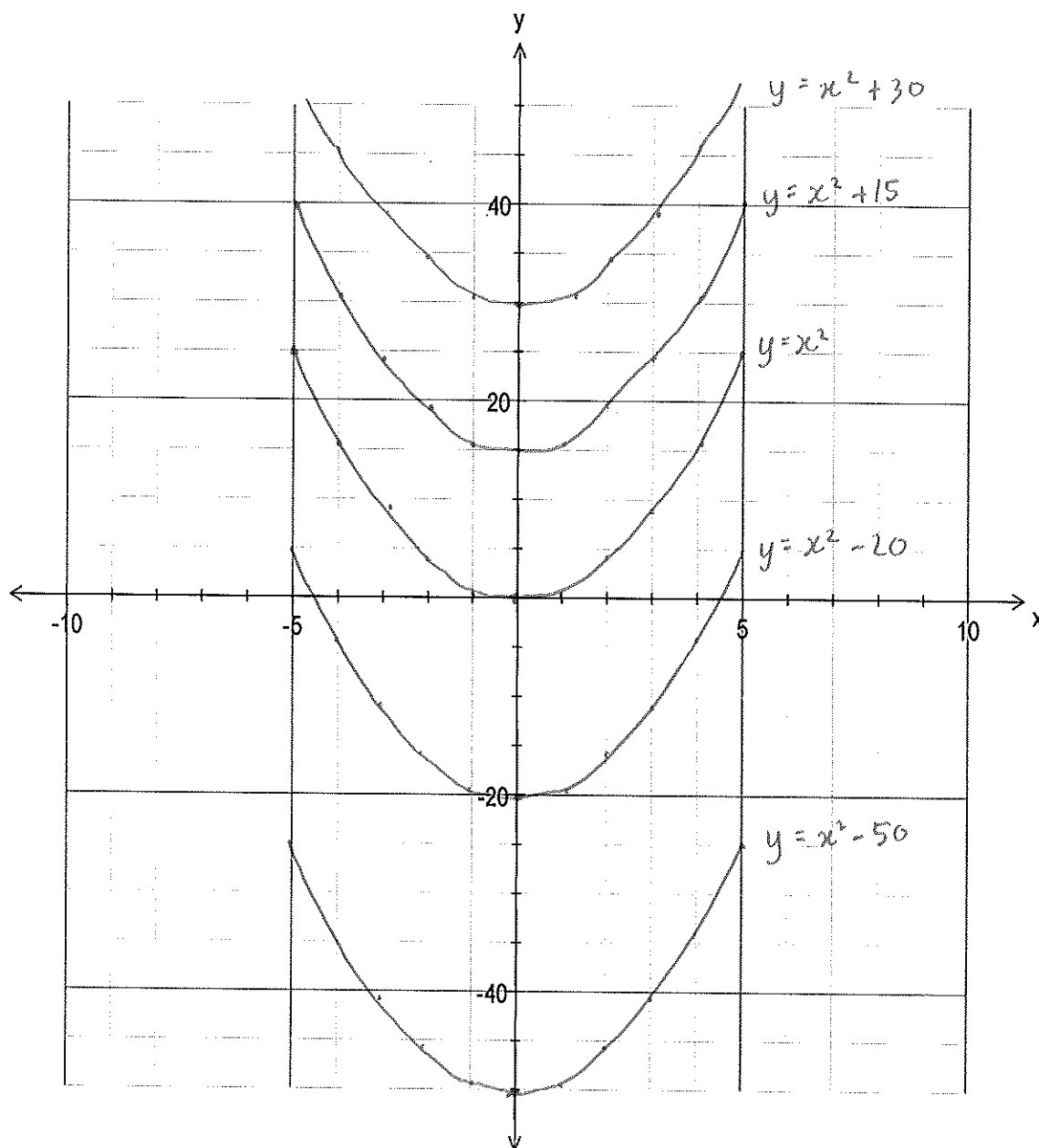
Negative a flips it over the x-axis

3.

a) Complete the table of values for the equations shown.

	-5	-4	-3	-2	-1	0	1	2	3	4	5
$y = x^2$	25	16	9	4	1	0	1	4	9	16	25
$y = x^2 + 15$	40	31	24	19	16	15	16	19	24	31	40
$y = x^2 + 30$	X	46	39	34	31	30	31	34	39	46	X
$y = x^2 - 20$	5	-4	-11	-16	-19	-20	-19	-16	-11	-4	5
$y = x^2 - 50$	-25	-34	-41	-46	-49	-50	-49	-46	-41	-34	-25

b) Now draw each of the equations on the axis below, be sure to label each equation.



c) Looking at the graphs you have drawn in the form $y = x^2 + c$, describe how the value of "c" changes (transforms) the original equation $y = x^2$?

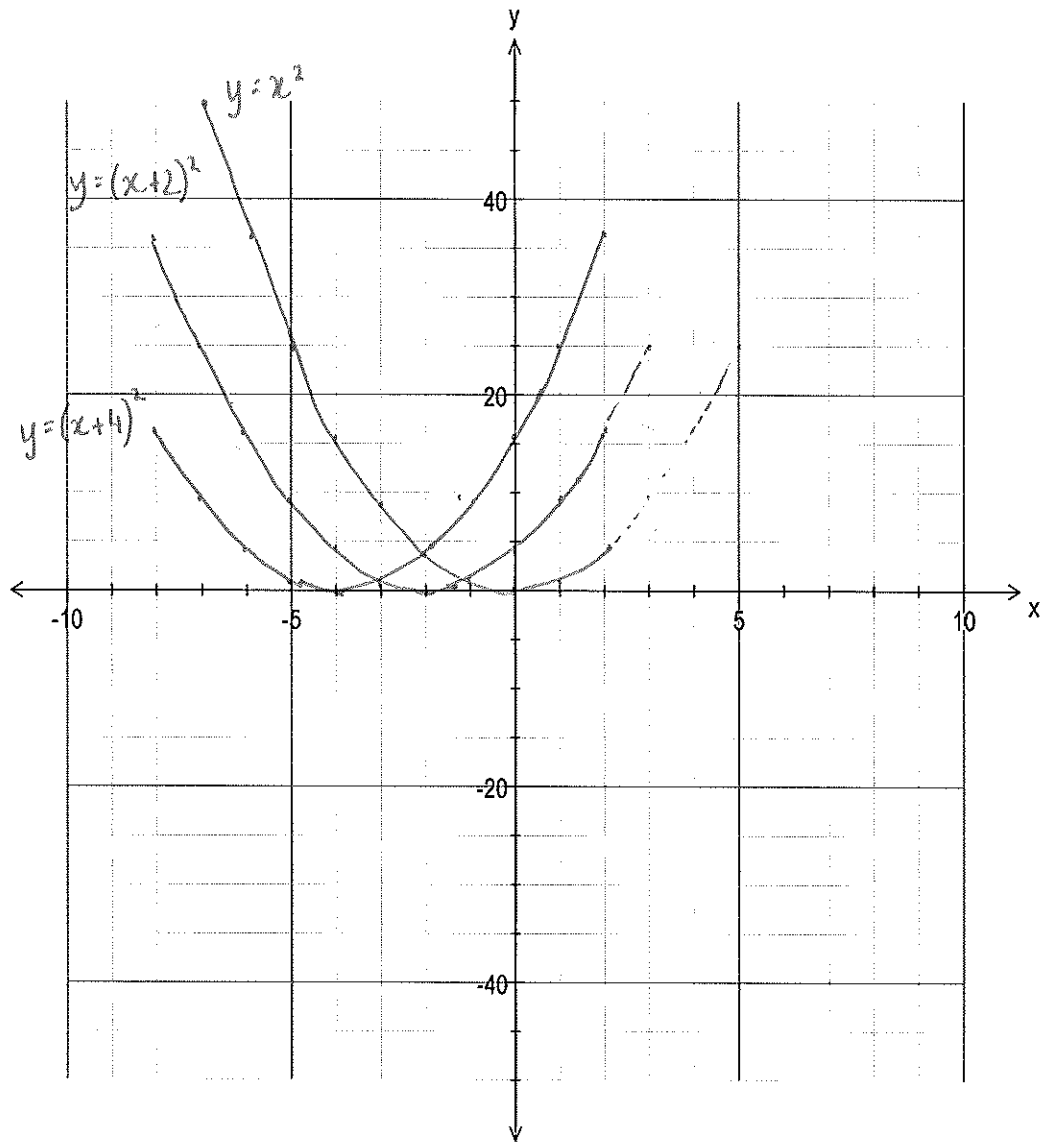
+c moves up
-c moves down

4.

a) Complete the table of values for the equations shown.

	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2
$y = x^2$		49	36	25	16	9	4	1	0	1	4
$y = (x + 2)^2$	36	25	16	9	4	1	0	1	4	9	16
$y = (x + 4)^2$	16	9	4	1	0	1	4	9	16	25	36

b) Now draw each of the equations on the axis below, be sure to label each equation.



c) Looking at the graphs you have drawn in the form $y = (x + b)^2$, describe how the value of "b" changes (transforms) the original equation $y = x^2$?

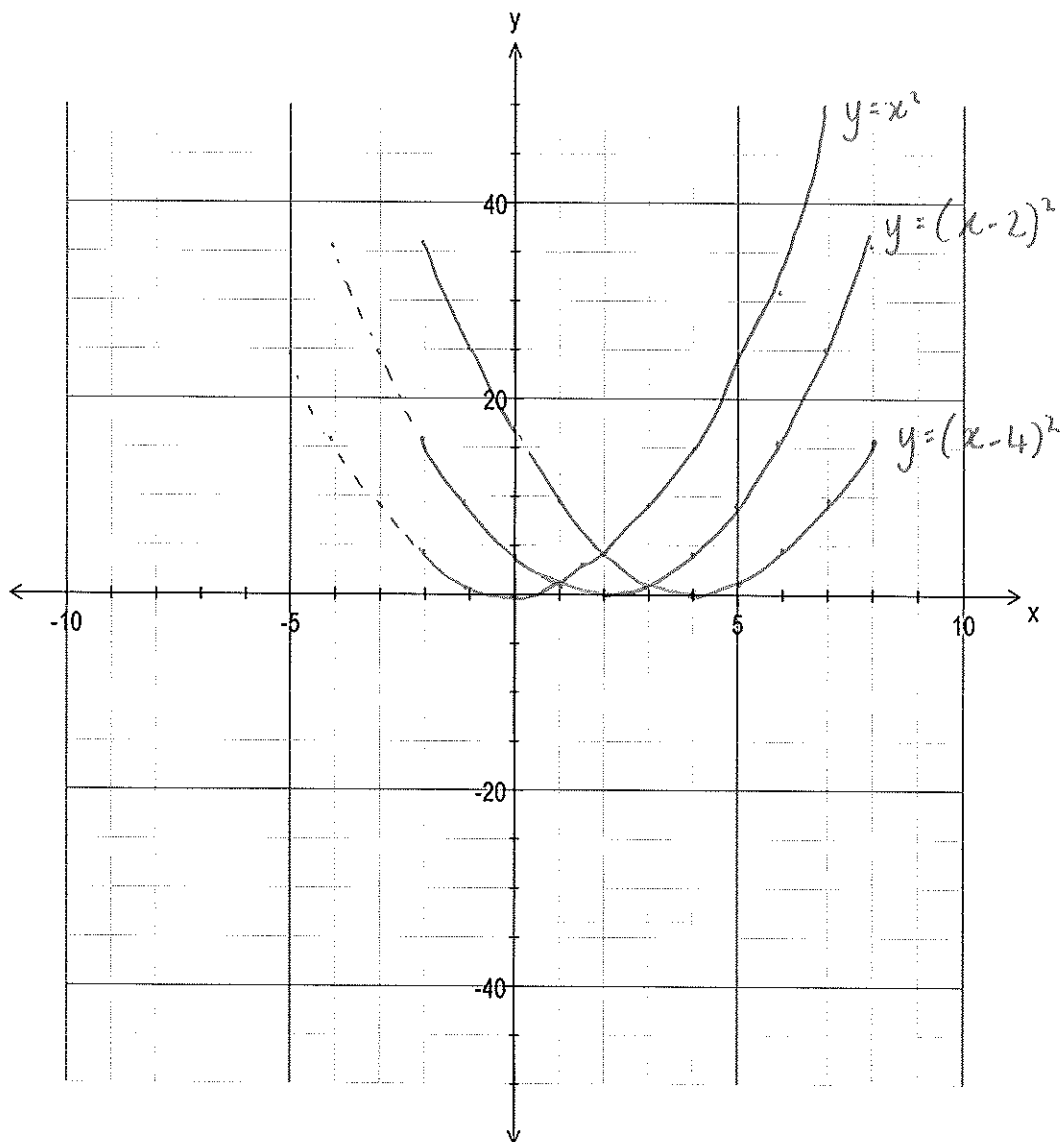
The graph moves to the left

5.

a) Complete the table of values for the equations shown.

	-2	-1	0	1	2	3	4	5	6	7	8
$y = x^2$	4	1	0	1	4	9	16	25	36	49	X
$y = (x - 2)^2$	16	9	4	1	0	1	4	9	16	25	36
$y = (x - 4)^2$	36	25	16	9	4	1	0	1	4	9	16

b) Now draw each of the equations on the axis below, be sure to label each equation.



c) Looking at the graphs you have drawn in the form $y = (x - b)^2$, describe how the value of "b" changes (transforms) the original equation $y = x^2$?

The graph moves to the right.